



UNIVERSITY OF
ZULULAND

FACULTY OF SCIENCE, AGRICULTURE AND ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

ASSIGNMENT 2

4CPS312: Distributed Systems Development

Collaborative Self-Study Assignment

Each group must independently study the complete topic of Processes in Distributed Systems and use the supplied prompts to develop a shared understanding of processes and threads, context switching and concurrency, virtualization and containers, clients and servers, stateful and stateless servers, server clusters, and code migration and mobility.

Instructions

- Watch the recorded lectures provided for the complete topic.
- Study the accompanying slides and prescribed textbook section.
- Prepare individual notes before participating in the group activity.
- Contribute to the discussion of every part of the assignment.
- Every student remains responsible for the complete topic. The work should not be divided in a way that results in each member studying only one section.
- **Due Date: Monday, 7 September 2026**
- Each group must submit **one consolidated group assignment**: Only one designated group member should upload the group assignment.
- The group assignment must include the names and student numbers of all group members and a contribution statement explaining each member's contribution.
- In addition, **each student must submit an individual reflection** describing their learning experience and personal contribution to the group activity.
- The group assignment and individual reflections must be submitted electronically through the **respective submission links provided on the LMS**:
- Every student is responsible for uploading their own individual reflection.
- Submissions sent by email or uploaded through another student's account will not be accepted.
- All submissions must be completed by the stated deadline.

Scenario

A university is developing a distributed online learning platform that supports:

1. Student login and access to course content
2. Live and recorded lectures
3. Online assessments and automated marking
4. File uploads and downloads
5. Simultaneous access by many students
6. Deployment across multiple servers
7. Recovery following server or network failure
8. Access from different devices and locations

Your group has been asked to recommend how the processes supporting the platform should be organised, deployed and managed.

Use the following prompts to guide your group discussion and assignment.

Part A: Processes, Threads and Context Switching

Consider a platform server that must receive requests, communicate with other services, access databases and return responses to several students.

Discuss and respond to the following:

1. Distinguish between a process and a thread.
2. Explain what resources are maintained by a process and what resources are shared by threads within the same process.
3. Explain what information must be saved during a context switch.
4. Explain why switching between threads is generally less expensive than switching between processes.
5. Explain why a distributed server may use multiple threads instead of creating a separate process for every request.
6. Identify the risks associated with allowing multiple threads to share memory.
7. Identify circumstances in which separate processes would be preferable to multiple threads.

Group decision: Recommend whether the platform's server application should use: A single process with multiple threads, Multiple single-threaded processes, Multiple multithreaded processes or another justified arrangement. Explain your decision.

Part B: Concurrency in Clients and Servers

A student's browser may simultaneously download text, images and video. At the same time, the platform server receives requests from many students.

Discuss and respond to the following:

1. Explain how multithreading can help a client hide communication delays.
2. Explain how multithreading can improve server responsiveness.
3. Distinguish between an iterative server and a concurrent server.
4. Explain what happens when an iterative server performs a slow or blocking operation.
5. Explain how the dispatcher–worker model handles incoming requests.
6. Determine whether increasing the number of threads will always improve performance.
7. Identify the resources that may limit the number of requests handled concurrently.
8. Identify one concurrency problem that may occur when several students submit assessments simultaneously.

Group decision: Recommend an appropriate server model for the learning platform. Explain how it would handle several simultaneous assessment submissions and how concurrency would be controlled.

Part C: Virtualization, virtual machines and containers

The university wants to deploy several services on shared physical infrastructure. The services must be isolated, portable and independently managed.

Discuss and respond to the following:

1. Explain the principle of virtualization and its relevance to distributed systems.
2. Distinguish between a process virtual machine, native virtual machine monitor and hosted virtual machine monitor.
3. Compare virtual machines and containers in terms of isolation, resource use, portability and startup time.
4. Explain how namespaces and control groups support containers.
5. Recommend how the learning platform’s authentication, content and assessment services should be deployed.
6. Justify whether the services should use virtual machines, containers or a combination.
7. Determine whether virtualization completely prevents services on the same physical host from interfering with one another.

Group decision: Recommend whether the authentication, content and assessment services should be deployed using: Virtual machines, Containers, Containers inside virtual machines, or a justified combination of these approaches. Support the recommendation using isolation, performance, portability, scalability and recovery considerations.

Part D - Clients and Servers

The platform is accessed through web browsers and mobile applications that communicate with several server-side services.

1. Discuss and respond to the following:

2. Identify the main responsibilities of the client.
3. Identify the main responsibilities of the server.
4. Explain how client-side software can support access, location, migration, replication and failure transparency.
5. Explain how a client locates and contacts the required server.
6. Explain the role of server ports and service endpoints.
7. Determine what could go wrong if too much application logic is placed on the client.
8. Determine what could happen if almost all processing is placed on the server.
9. Explain how a web browser functions as a client in a distributed system.

Group decision: Recommend an appropriate division of responsibilities between the student application and the platform's servers. Justify the proposed division.

Part E - Stateful and stateless servers

A student begins an online assessment, answers several questions and temporarily loses network connectivity.

1. Discuss and respond to the following:
2. Distinguish between a stateful and a stateless server.
3. Identify the information that a stateful assessment server might retain about the student.
4. Explain the advantages of a stateless server design.
5. Explain the performance benefits that a stateful design may provide.
6. Explain what may happen to client state when a stateful server fails.
7. Determine whether a service can be stateless while using a stateful database.
8. Explain how a student could resume an assessment following temporary disconnection.
9. Explain how the choice between stateful and stateless design affects server replication and load balancing.

Group decision: Determine whether the assessment service should be stateful or stateless. Explain where the assessment state should be stored and how it should be recovered following failure.

Part F - Server Clusters and Request Distribution

The platform experiences a sudden increase in traffic when examination results are released.

1. Discuss and respond to the following:
2. Explain why a single server may be unable to support the increased demand.
3. Explain how a server cluster can improve scalability and availability.
4. Explain the role of a request dispatcher or load balancer.
5. Explain how a request dispatcher could become a bottleneck or single point of failure.
6. Explain how the Domain Name System may assist with request distribution.
7. Explain how users may be directed towards geographically closer servers.

8. Explain how caching can reduce the workload of the origin server.
9. Identify one consistency problem that may result from caching or replication.

Group decision: Propose a request-distribution strategy for supporting increased demand. The proposal must address scalability, availability, bottlenecks and consistency.

Part G - Code Migration and Mobility

The platform must process a large collection of student submissions stored at a remote location. Transferring all the data may be expensive and may raise privacy concerns.

Discuss and respond to the following:

1. Explain the main reasons for migrating code in a distributed system.
2. Explain why moving computation closer to data may reduce communication.
3. Distinguish between code shipping and code fetching.
4. Distinguish between weak and strong mobility.
5. Explain what is transferred under weak mobility.
6. Explain what additional information is transferred under strong mobility.
7. Distinguish between migration and cloning.
8. Explain why strong mobility is difficult across heterogeneous systems.
9. Explain how virtual machines can support service migration.
10. Identify the security and privacy risks of executing migrated code.

Group decision: Determine whether the automated marking service should: Move the data to the marking code, Move the code closer to the data, Distribute the computation across locations OR Use another justified approach.

Discuss the performance, privacy, security and portability implications of the decision.

Part H - Integrated Design

Using the conclusions from Parts A-E, produce an integrated design for the online learning platform. The group must specify:

1. The main client and server processes.
2. Which components will use multiple threads.
3. How concurrent requests will be handled.
4. Which services will be stateful or stateless.
5. Where application state will be stored.
6. Which components will run in virtual machines or containers.
7. How requests will be distributed across servers.
8. How the system will respond to server failure.
9. Where code or computation may be migrated.
10. The main performance, scalability, reliability, portability, privacy and security trade-offs.

Include one clearly labelled architecture diagram.

Submission Requirements

Submit one group report of approximately 6-10 pages, excluding the cover page, references and appendices. The submission must contain:

1. Cover page
2. Introduction
3. Responses to Parts A-E
4. Integrated design and architecture diagram
5. Discussion of major design trade-offs
6. Conclusion
7. References
8. Group contribution statement

Students should not copy definitions directly from the slides. Marks will be awarded for explanation, comparison, application and justification.

Group Contribution Statement

Include a contribution table that identifies the specific work completed by each member. General statements such as 'participated in the assignment' are insufficient.

Group member	Specific contribution	Sections or outputs completed

Group Assignment Rubric

Criterion	Marks
Understanding of processes, threads and concurrency	12
Understanding of virtualization and containers	12
Analysis of client and server organisation	12
Analysis of stateful and stateless server design	12
Analysis of server state, clustering and request distribution	12
Understanding and application of code migration	10
Integrated design and architecture diagram	12
Analysis of design trade-offs	6
Quality of justification and analysis of trade-offs	6

Criterion	Marks
Organisation, referencing and contribution statement	6
Total	100

Individual Quiz

After completing the group assignment, each student will write an individual quiz covering the complete topic. The quiz may assess:

1. Processes and threads
2. Context switching and concurrency
3. Virtualization and containers
4. Client and server organisation
5. Stateful and stateless servers
6. Server clusters and request distribution
7. Code migration and mobility
8. Application of these concepts to distributed-system scenarios

The group assignment assesses collaborative application and analysis, while the quiz assesses each student's individual understanding of the complete topic.

Although the assignment is completed in groups, every student is responsible for studying the complete topic. The quiz may assess any section of the recorded lectures, supporting slides and prescribed readings.